

Enzyme Docking & Visualization Project

Overview

This project performs:

- **Molecular docking** of an enzyme and substrate using **AutoDock Vina**.
 - **Visualization** of docking results using **PyMOL**.
 - Optional **AlphaFold structure prediction** (via ColabFold or local setup).
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Folder Structure

```
1  Enzyme_project/  
2  └─ data/                # Input files (enzyme structure,  
ligand)  
3  │   └─ predicted_structure.pdbqt  
4  │   └─ ligand.pdbqt  
5  │   └─ esterase.fasta   # For AlphaFold predictions  
6  └─ results/            # Output files (docked ligand, images)  
7  │   └─ docked.pdbqt  
8  │   └─ EstB_binding.png  
9  └─ scripts/            # Python scripts  
10 │   └─ Docker_substrate.py  
11 │   └─ run_alphafold.py  
12 │   └─ visual.py  
13 │   └─ test.py  
14 └─ requirements.txt  
15
```

Scripts Explained

1. Docker_substrate.py

- **Purpose:** Runs AutoDock Vina docking.
- **Inputs:**
 - data/predicted_structure.pdbqt (enzyme)
 - data/ligand.pdbqt (substrate)
- **Output:**
 - results/docked.pdbqt (docked ligand)
- **Run:**

```
1 python scripts/Docker_substrate.py
2
```

2. run_alphafold.py

- **Purpose:** Visualizes docking results in PyMOL.
- **Inputs:**
 - data/predicted_structure.pdbqt
 - results/docked.pdbqt
- **Output:**
 - results/EstB_binding.png (rendered image)
- **Run:**

```
1 python scripts/run_alphafold.py
2
```

- **Note:** Requires **PyMOL** installed (GUI or open-source).
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3. visual.py

- **Purpose:** General visualization of enzyme-ligand complex.
- **Inputs:**
 - Protein (PDB)
 - Ligand (PDBQT)
- **Output:**
 - results/complex.png
- **Run:**

```
1 python scripts/visual.py
2
```

4. test.py

- **Purpose:** Tests PyMOL installation and rendering.
- **Output:**
 - results/test_output.png
- **Run:**

```
1 python scripts/test.py
2
```

How Scripts Are Connected

1. **Docker_substrate.py** → Generates docking result (docked.pdbqt).
 2. **run_alphafold.py** → Uses docking result + enzyme structure to create visualization.
 3. **visual.py** → Alternative visualization script.
 4. **test.py** → Confirms PyMOL works before running visualization scripts.
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Environment Setup

- **Python 3.8+**
- Install dependencies:

```
1 pip install -r requirements.txt
2
```

- For PyMOL:
 - **Windows:** Install official PyMOL GUI → <https://pymol.org/2/>
 - **Linux:**

```
1 conda install -c schrodinger pymol
2
```

```
1 - Or use:
```

```
1 pip install pymol-open-source
2
```

AlphaFold Prediction

- Use **ColabFold** for easy setup:



<https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2/mmseqs2.ipynb>

- Upload esterase.fasta and run all cells.
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Notes

- Docking and visualization can run on CPU.
- AlphaFold predictions require GPU (use ColabFold if no local GPU).